

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1921

Steamboat 138 kV
November 24, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1921. The Interconnection Customer proposes a 247.4 MW battery energy storage facility interconnecting at the Steamboat 138 kV substation in the Virginia area. The request was received on October 31, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 657 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,771,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Laysmith Hydro - Leesville 138 kV	Huntington Court Substation - Reusens 345 kV	173.1	216	13.4
Poages Mill - Floyds Knos Substation 138 kV	Safeway Stores - Asheville Substation 138 kV	167.7	223	13.2
Steamboat - Muddy Ranch 138 kV	Huntington Court - Steamboat 138 kV	135.2	229	100.0
Hinton Ap - Oxnard 138 kV	Centrl Juncos 3202 - Colton 138 kV L	126.3	328	9.2
Dekalb - Oxnard 138 kV	Colton - Central Juncos 3202 138 kV	125.3	337	9.2
Banning - Franklin-Elk Grove Substation 138 kV	S Texa - Franklin-Elk Grove Substation 138 kV	124.8	354	6.1
De Kalb - McClain Energy Facility 138 kV line	Colton - Central Juncos 3202 138 kV	124.5	344	9.2
Banning - Deert Outpost Substation 138 kV	S Texa - Franklin-Elk Grove 138 kV	123.3	369	6.1
Floyds Knobs - Hancock 138 kV L	Asheville - Safeway Stores 138 kV	122.8	213	14.3
Littlefork - Lake Lure 138 kV	Louisa Generation Facility - Fluvanna P.S 345 kV	116.8	918	6.0
Franklin-Elk Grove - New London 138 kV Line	Bear - Clover Substation 345 kV	116.5	637	9.0
Steamboat - Huntington Court 138 kV	Muddy Ranch - Steamboat 138 kV line	110.7	280	100.0
Franklin-Elk Grove - South Texa 138 kV	Desert Outpost Substation - S Texa 138 kV	110.1	611	7.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W Bassett Substation - Ladysmith Hydro 138 kV #1	W Bassett - Ladysmith Hydro 138 kV Line 2	107.1	323	11.8
Desert Outpost - S Texa 138 kV Line	Franlin-Elk Grove - S Texa 138 kV Line	106.9	815	9.0
Candelaria - Monel 138 kV	Bear - Clover 345 kV	104.7	633	5.1
W Bassett - Ladysmith Hydro 138 kV line	Ladysmith Hydro - W Bassett 138 kV	102.8	336	11.8
Reach - Candelaria 138 kV Line	Bear - Clover 345 kV Line	102.7	631	5.1
Desert Outpost - Monel 138 kV L	Franklin-Elk Grove - New London 138 kV line	102.2	557	5.8
Chancellor Row - Bear Island 345 kV Line	Leavitt Substation - Birchwood 345 kV line	101.9	3,158	14.0
Bluff City - Lake Lure 138 kV	Louisa Generation Facility - Fluvanna P.S 345 kV	101.2	974	5.8
Huntington Court 345-138 kV XFMR	Lone Tree - Craven County Wood Energy Lp 138 kV line	101.2	430	29.6
Bear - Leavitt 345 kV Cir. 2	Bear - Leavitt 345 kV Circuit 1	101.2	2,822	15.5
Haggin - Mitchell Dam 345 kV	Bear - Mecklenburg Power Sta 345 kV L	100.5	1,962	6.0
Bear - Mecklenburg Power Sta 345 kV	Bear - Clover 345 kV	100.1	3,978	20.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Laysmith Hydro - Leesville 138 kV	Rebuild line with higher-capacity conductor	\$59,811,000	22% (\$12,861,000)
Poages Mill - Floyds Knos Substation 138 kV	Rebuild line with higher-capacity conductor	\$4,269,000	30% (\$1,273,000)
Steamboat - Muddy Ranch 138 kV	Rebuild line with higher-capacity conductor	\$4,274,000	65% (\$2,793,000)
Hinton Ap - Oxnard 138 kV	Rebuild line with higher-capacity conductor	\$5,911,000	5% (\$287,000)
De Kalb - Oxnard 138 kV	Rebuild line with higher-capacity conductor	\$5,206,000	5% (\$253,000)
Banning - Franklin-Elk Grove Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,857,000	5% (\$90,000)
De Kalb - McClain Energy Facility 138 kV line	Rebuild line with higher-capacity conductor	\$5,561,000	5% (\$270,000)
Banning - Desert Outpost 138 kV L	Rebuild line with higher-capacity conductor	\$12,562,000	5% (\$609,000)
Floyds Knobs - Hancock 138 kV L	Rebuild line with higher-capacity conductor	\$3,293,000	32% (\$1,059,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Littlefork - Lake Lure 138 kV	Rebuild line with higher-capacity conductor	\$9,626,000	3% (\$278,000)
Franklin-Elk Grove - New London 138 kV Line	Rebuild line with higher-capacity conductor	\$3,191,000	4% (\$139,000)
Steamboat - Huntington Court 138 kV	Rebuild line with higher-capacity conductor	\$2,548,000	65% (\$1,665,000)
Franklin-Elk Grove - South Texa 138 kV	Rebuild line with higher-capacity conductor	\$15,059,000	5% (\$713,000)
W Bassett Substation - Ladysmith Hydro 138 kV #1	Reconductor line	\$3,674,000	28% (\$1,013,000)
Desert Outpost - S Texa 138 kV	Reconductor line	\$620,000	5% (\$29,000)
Candelaria - Monel 138 kV	Reconductor line	\$14,209,000	6% (\$889,000)
West Bassett - Ladysmith Hydro Substation 138 kV #2	Reconductor line	\$5,130,000	28% (\$1,415,000)
Candelaria - Reach 138 kV	Reconductor line	\$6,420,000	6% (\$401,000)
Monel - Desert Outpost 138 kV	Reconductor line	\$1,610,000	5% (\$84,000)
Chancellor Row - Bear Island Substation 345 kV line	Reconductor line	\$21,474,000	3% (\$590,000)
Bluff City - Lake Lure 138 kV	Reconductor line	\$6,863,000	4% (\$258,000)
Huntington Court 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$12,964,000	16% (\$2,079,000)
Bear - Leavitt 345 kV Ckt 2	Reconductor line	\$57,078,000	3% (\$1,648,000)
Haggin - Mitchell Dam Substation 345 kV	Reconductor line	\$6,490,000	3% (\$171,000)
Bear - Mecklenburg Power Sta 345 kV	Reconductor line	\$32,643,000	3% (\$904,000)
	Total allocated to project		\$31,771,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.